Tut-3

Microprocessor Programming and Interfacing

Q . Discuss various directives : equ, db, dw, dd, dup, align

```
COUNT     EQU  10
CONST     EQU  20H

          MOV  AH, COUNT
          MOV  AL, CONST
          ADD  AH,AL

RES       DB        10DUP(?)
```

DB, DW, DD

```
DATA1     DB   45H, 35H, 74H
DATA2     DW   2000H, 37H, 2222H
DATA3     DD   234567ABH
```

```
.Model Tiny

.data
dat1    db      'a'
        align   2
dat2    db      'b'
.code
.startup
        mov     al,dat1
        add     dat2,al
.exit
end
```

0000	31	0010	00
0001	32	0011	45
0002	33	0012	23
0003	34	0013	00
0004	35	0014	03
0005	36	0015	00
0006	67	0016	00
0007	66	0017	FF
0008	01	0018	FF
0009	02	0019	FF
000A	03	001A	FF
000B	61	001B	FF
000C	F0	001C	FF
000D	0C	001D	FF
000E	00	001E	FF
000F	0D	001F	FF

```
        ORG   0000H
MSG2    DB    '123456'
MSG3    DW    6667H
data1   db    1,2,3
        db    'a'
        db    11110000b

data2   dw    12,13
        dw    2345H
        dd    300H
        DB    9 DUP(FFH)
```

Q . Now, given below is an 8086 assembly program segment.

```
.model tiny
.data
first dw 2345h,4h,34h
second db 12h
third dd 12345678h,0AABBCCDDh
align 2
dat1 db 0ah,0bh
dat2 equ 23h
align 2
dat5 db 12h,34h,56h,67h,0ah
dat6 db 34h,12h,23h,21h,45h
dat7 db 5 dup(?)
.code
.startup
```

Assuming that the data declarations are stored in memory locations starting from DS: 0180, write the contents of memory that result from data declarations in the program.

DS:0180								
DS:0188								
DS:0190								
DS:0198								

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align 2
dat1 db 0ah,0bh
dat2 equ 23h
align 2
dat5 db 12h,34h,56h,67h,0ah
dat6 db 34h,12h,23h,21h,45h
dat7 db 5 dup(?)
.code
.startup
```

DS:0180	45	23	04	00	34	00	12	78
DS:0188	56	34	12	DD	CC	BB	AA	00
DS:0190	0A	0B	12	34	56	67	0A	34
DS:0198	12	23	21	45	00	00	00	00

Q For the data declaration given in Q1, write an ALP to perform byte- addition as shown below

dat5	12h	34h	56h	67h	0ah
	+	+	+	+	+
dat6	34h	12h	23h	21h	45h
dat7	46h	46h	79h	88h	4Fh

The result of addition should be stored in corresponding location in dat7.

The data inputs for addition (in dat5,dat6 are such that the addition will not result in a carry)

```
.model tiny
.data
first dw 2345h,4h,34h
second db 12h
third dd 12345678h,0AABBCCDDh
align 2
dat1 db 0ah,0bh
dat2 equ 23h
align 2
dat5 db 12h,34h,56h,67h,0ah
dat6 db 34h,12h,23h,21h,45h
dat7 db 5 dup(?)
.code
.startup
```

```
dat5 db 12h,34h,56h,67h,0ah
dat6 db 34h,12h,23h,21h,45h
dat7 db 5 dup(?)

.code
.startup
        mov si,offset dat5
        mov di,offset dat6
        mov bx, offset dat7
        mov cl,5
l1:     mov al,[si]
        mov ah,[di]
        add al,ah
        mov [bx],al
        inc si
        inc di
        inc bx
        dec cl
        jnz l1
.exit
end
```

Difference between lea & mov offset based access to be explained

.model tiny
.386
.data
array1 db $0a_h$,bc_h ,de_h ,$0f5_h$,11_h, 56_h ,78_h ,$0ff_h$,$0ff_h$,23_h 4ah, …
array2 db 50 dup(0)
.code
startup
mov cx,32_h
lea si,array1
lea di,array2